Integral limit theorems

REAL VARIABLES I

Fall 2026

We have already established the monotone convergence theorem, which allows the exchange of limits and integrals for non-decreasing sequences of non-negative functions: $0 \leq f_n \uparrow f$ implies $\int_X f_n \, \mathrm{d}\mu \to \int_X f \, \mathrm{d}\mu$. In many applications, however, sequences of functions oscillate, take both positive and negative values, and lack any monotonicity. We wish to identify general conditions under which pointwise convergence almost everywhere guarantees convergence of integrals without requiring monotonicity.

Pointwise convergence alone does not suffice. As we show in counterexamples, convergence of integrals may fail because "mass escapes to infinity" or "concentrates into a vanishing set". The Lebesgue dominated convergence theorem precludes this behavior by requiring the sequence to be bounded by a single integrable function, called a dominating profile.

We will first prove dominated convergence using Fatou's lemma. We will then use Egorov's theorem to prove the more general Vitali convergence theorem: tightness prevents mass from escaping, and equi-uniform integrability prevents concentration. Dominated convergence follows by verifying these two properties for a sequence with an integrable dominating profile.

After studying these notes, you will be able to do the following.

- State the Lebesgue dominated convergence theorem and contrast its hypotheses with the monotone convergence theorem.
- Construct sequences showing how limit-integral exchange fails when mass escapes to infinity or concentrates at a point, and analyze their minimal dominating profiles.
- Formulate and apply Fatou's lemma to sequences of non-negative measurable functions.
- State and prove Egorov's theorem using dyadic scales and explain why its finite-measure hypothesis is needed.
- Define tightness and equi-uniform integrability, prove Vitali's convergence theorem using Egorov's theorem, and deduce dominated convergence.

Throughout these notes, $(X, \mathcal{A}, \mu)$ denotes a measure space, and pointwise equalities and inequalities are understood to hold μ-almost everywhere unless stated otherwise.

9.1 Dominated convergence

Theorem 9.1 (Lebesgue dominated convergence theorem). Let $(X, \mathcal{A}, \mu)$ be a measure space. Suppose that $f_n \colon X \to \mathbb{R}$ are measurable functions that converge μ-almost everywhere. Assume there exists an integrable function $g \colon X \to [0, +\infty)$ such that

$$|f_n(x)| \leq g(x) \quad \text{for } \mu\text{-almost every } x \in X \text{ and for every } n \in \mathbb{N}.$$

Then the functions f_n and $f \coloneqq \lim_n f_n$ are integrable and,

$$\lim_{n\to\infty} \int_X f_n \, \mathrm{d}\mu = \int_X \lim_{n\to\infty} f_n \, \mathrm{d}\mu. \tag{9.1}$$

Furthermore, the stronger result

$$\lim_{n\to\infty} \int_X |f_n - f| \, \mathrm{d}\mu = 0, \qquad \text{where } f \coloneqq \lim_{n\to\infty} f_n. \tag{9.2}$$

holds

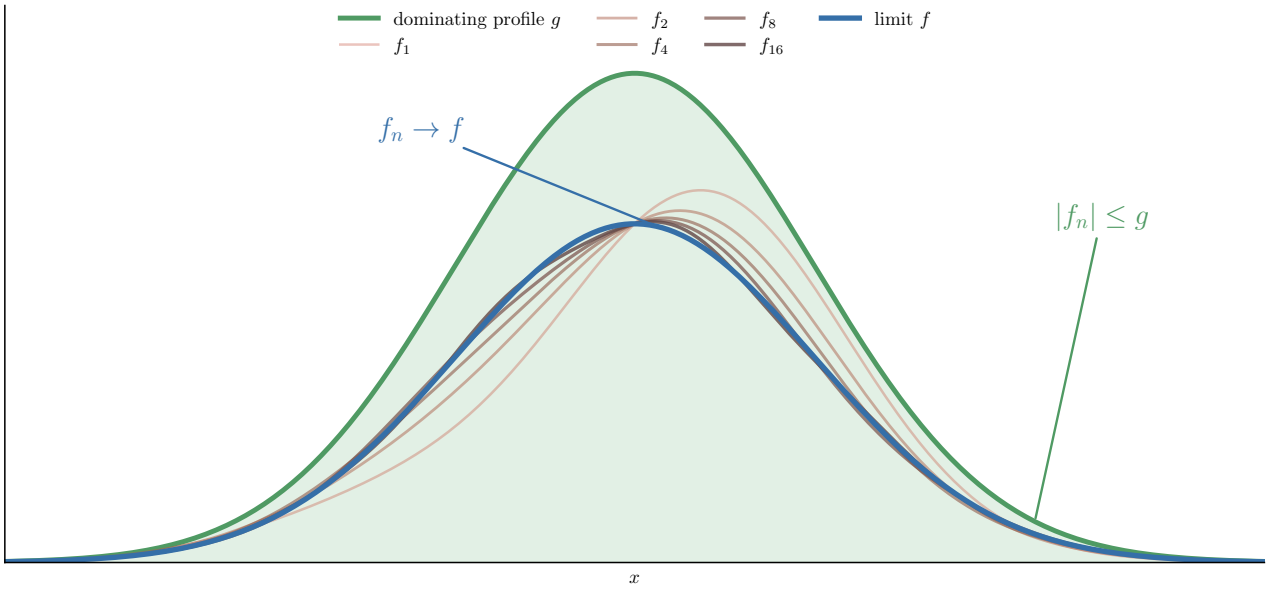


Figure 9.1: A sequence of functions f_n that converge to a limit function while remaining below one fixed dominating profile g.

Figure description: A plot of several increasingly similar curves f sub n converging to a blue limit curve f, all lying below a green shaded dominating profile g.

The monotone convergence theorem requires non-negative functions and the order relation $f_n \uparrow \lim_n f_n$, but it allows the limiting integral to be infinite. In contrast, the dominated convergence theorem applies to signed functions without any monotonicity assumption, but requires the sequence to be controlled by a single integrable bound g.

The second claim of Theorem 9.1 is indeed stronger than just the exchange of limit and integrals. This conclusion is indeed stronger than (9.1): because

$$\left| \int_X f_n \, \mathrm{d}\mu - \int_X \lim_{k\to\infty} f_k \, \mathrm{d}\mu \right| = \left| \int_X (f_n - f) \, \mathrm{d}\mu \right| \le \int_X |f_n - f| \, \mathrm{d}\mu,$$

which immediately implies (9.1).

When $(X, \mathcal{A}, \mu)$ is the counting measure on $\mathbb{N}$, the Lebesgue integral is an infinite series, and dominated convergence specializes to the following classical result.

Proposition 9.2 (Dominated convergence for series). Suppose that sequences of real numbers satisfy $a_n(k) \to a(k)$ as $n \to \infty$ for every $k \in \mathbb{N}$. If there exists a sequence $b(k) \ge 0$ such that $|a_n(k)| \le b(k)$ for all $n, k \in \mathbb{N}$ and $\sum_{k=1}^{+\infty} b(k) < +\infty$, then

$$\lim_{n\to\infty} \sum_{k=1}^{+\infty} |a_n(k) - a(k)| = 0 \quad \text{and} \quad \lim_{n\to\infty} \sum_{k=1}^{+\infty} a_n(k) = \sum_{k=1}^{+\infty} a(k).$$

Proof of Proposition 9.2. The pointwise limit satisfies $|a(k)| \leq b(k)$ for all $k \in \mathbb{N}$. Let $\varepsilon > 0$. Because $\sum_{k=1}^{\infty} b(k) < +\infty$, the series tail vanishes: choose an integer $K \in \mathbb{N}$ large enough that

$$\sum_{k=K+1}^{+\infty} b(k) < \varepsilon.$$

On the finite index set $\{1, \dots, K\}$, pointwise convergence $a_n(k) \to a(k)$ holds for each k. Hence there exists an index $N \in \mathbb{N}$ such that for all $n \geq N$ and each $k \in \{1, \dots, K\}$,

$$|a_n(k) - a(k)| < \frac{\varepsilon}{K}.$$

Using the triangle inequality $|a_n(k) - a(k)| \leq |a_n(k)| + |a(k)| \leq 2b(k)$ on the tail indices $k > K$, we obtain for all $n \geq N$:

$$\begin{aligned}\sum_{k=1}^{+\infty} |a_n(k) - a(k)| &= \sum_{k=1}^{K} |a_n(k) - a(k)| + \sum_{k=K+1}^{+\infty} |a_n(k) - a(k)| \\ &< K \cdot \frac{\varepsilon}{K} + \sum_{k=K+1}^{+\infty} 2b(k) \\ &< \varepsilon + 2\varepsilon = 3\varepsilon.\end{aligned}$$

Because $\varepsilon > 0$ was arbitrary, the quantity 3ε can be made arbitrarily small. This proves $\lim_{n\to\infty} \sum_{k=1}^{+\infty} |a_n(k) - a(k)| = 0$. The triangle inequality $|\sum_{k=1}^{\infty} a_n(k) - \sum_{k=1}^{\infty} a(k)| \leq \sum_{k=1}^{\infty} |a_n(k) - a(k)|$ then yields convergence of the series. □

We will present two proofs of the dominated convergence theorem. One of them will follow a strategy analogous to this proof for series. Before developing the proofs, we examine two canonical counterexamples showing why pointwise convergence alone does not imply convergence of integrals, and analyze why dominating profiles fail to exist in each setting.

Example 9.3 (Mass escaping to infinity). On $\mathbb{R}$ equipped with Lebesgue measure, consider the sequence of indicator functions

$$f_n \coloneqq \mathbb{1}_{[n,n+1]}, \qquad n \in \mathbb{N}.$$

For each fixed $x \in \mathbb{R}$, choosing an integer $N > x$ ensures that $x \notin [n, n+1]$ for all $n \geq N$. Therefore $f_n(x) \to 0$ pointwise everywhere on $\mathbb{R}$. Nevertheless, the integral of every term in the sequence is

$$\int_{\mathbb{R}} f_n(x)\, \mathrm{d}\mathcal{L}(x) = \mathcal{L}([n, n+1]) = 1 \qquad \text{for all } n \in \mathbb{N},$$

while the integral of the pointwise limit is $\int_{\mathbb{R}} 0\, \mathrm{d}\mathcal{L}(x) = 0$. Thus $\lim_{n\to\infty} \int_{\mathbb{R}} f_n \neq \int_{\mathbb{R}} \lim_{n\to\infty} f_n$.
Let us analyze why no integrable dominating function g can exist for this sequence. Any dominating profile $g\colon \mathbb{R} \to [0, +\infty]$ must satisfy $g(x) \geq |f_n(x)| = \mathbb{1}_{[n,n+1]}(x)$ for all $n \in \mathbb{N}$ and almost every $x \in \mathbb{R}$. The minimal candidate dominator is the pointwise supremum

$$g^*(x) \coloneqq \sup_{n\in\mathbb{N}} |f_n(x)| = \sup_{n\in\mathbb{N}} \mathbb{1}_{[n,n+1]}(x) = \mathbb{1}_{[1,+\infty)}(x).$$

Any dominating function g must satisfy $g(x) \geq g^*(x)$ almost everywhere. Computing the integral of g^* gives

$$\int_{\mathbb{R}} g^*(x)\, \mathrm{d}\mathcal{L}(x) = \int_1^{+\infty} 1\, \mathrm{d}\mathcal{L}(x) = +\infty.$$

Alternatively, because the intervals $[n, n+1)$ are pairwise disjoint,

$$\int_{\mathbb{R}} g(x)\, \mathrm{d}\mathcal{L}(x) \geq \sum_{n=1}^{\infty} \int_{n}^{n+1} f_n(x)\, \mathrm{d}\mathcal{L}(x) = \sum_{n=1}^{\infty} 1 = +\infty.$$

Hence, no integrable dominating function exists. Here, integral convergence fails because the mass of f_n "escapes to infinity" along the real line.

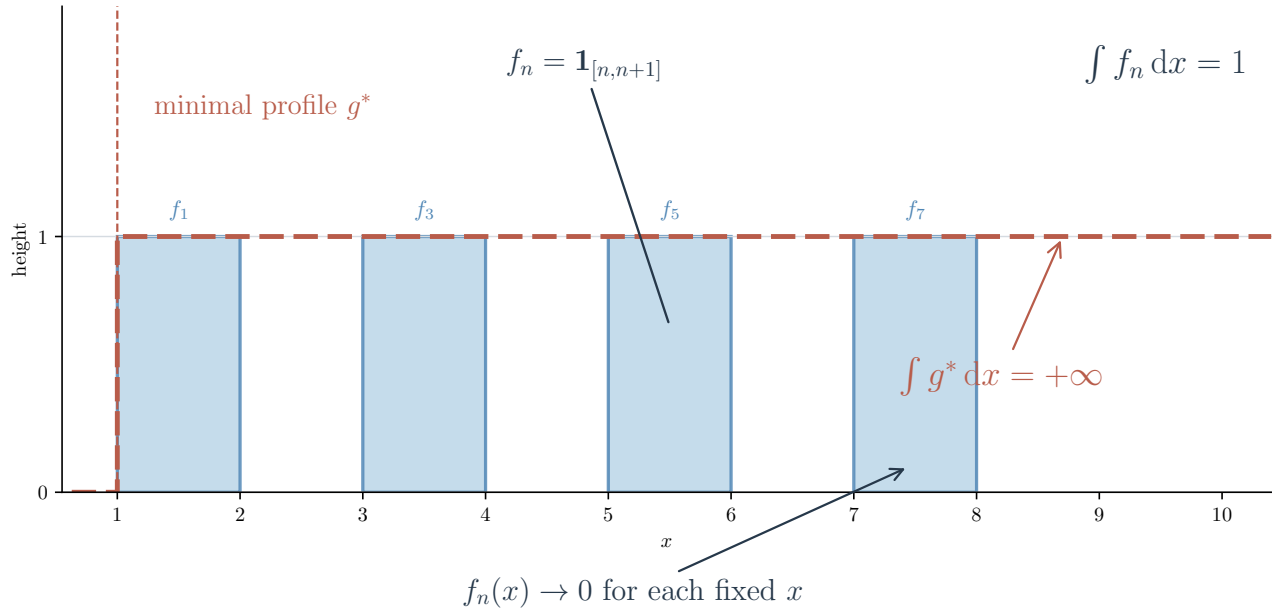


Figure 9.2: The unit-mass wave moves to the right and converge pointwise to zero. The minimal dominating profile is non-integrable: $\mathbf{1}_{[1,\infty)}$.

Figure description: A plot shows four separated blue unit-height pulses moving to the right, labelled f one, f three, f five, and f seven. A red dashed line and shaded tail show the minimal profile equal to one from x equals one onward, whose integral is infinite.

Example 9.4 (Mass concentrating at zero). On $\mathbb{R}$ with Lebesgue measure, define

$$f_n(x) \coloneqq 2^n \mathbb{1}_{[2^{-n}, 2^{1-n})}(x), \qquad n \in \mathbb{N}.$$

The intervals $I_n \coloneqq [2^{-n}, 2^{1-n})$ are pairwise disjoint and satisfy $\bigcup_{n=1}^{\infty} I_n = (0, 1]$. If $x \leq 0$ or $x > 1$, then $f_n(x) = 0$ for all n. If $x \in (0, 1]$, then x belongs to exactly one interval I_{n_0}. For all $n \neq n_0$, $f_n(x) = 0$. Consequently, $\lim_{n\to\infty} f_n(x) = 0$ for every $x \in \mathbb{R}$. However, the height 2^n and width 2^{-n} balance:

$$\int_{\mathbb{R}} f_n(x)\, \mathrm{d}\mathcal{L}(x) = 2^n \mathcal{L}(I_n) = 2^n (2^{1-n} - 2^{-n}) = 2^n \cdot 2^{-n} = 1 \qquad \text{for all } n \in \mathbb{N}.$$

Thus the limit of the integrals is 1, whereas the integral of the limit is 0.
Let us determine the minimal dominating profile $g^*(x) \coloneqq \sup_{n\in\mathbb{N}} f_n(x)$. Because the intervals I_n are pairwise disjoint,

$$g^*(x) = \sum_{n=1}^{\infty} 2^n \mathbb{1}_{[2^{-n}, 2^{1-n})}(x).$$

For $x \in [2^{-n}, 2^{1-n})$, we have $x < 2^{1-n}$, which gives $2^n > 1/x$. Thus $g^*(x) > \frac{1}{x}\mathbb{1}_{(0,1]}(x)$, matching the singularity of $1/x$ near 0. Integrating g^* over $\mathbb{R}$ yields

$$\int_{\mathbb{R}} g^*(x)\, \mathrm{d}\mathcal{L}(x) = \sum_{n=1}^{\infty} \int_{2^{-n}}^{2^{1-n}} 2^n \, \mathrm{d}\mathcal{L}(x) = \sum_{n=1}^{\infty} 1 = +\infty.$$

Every dominating function g satisfies $g \geq g^*$ almost everywhere, so $\int_{\mathbb{R}} g \, \mathrm{d}\mathcal{L}(x) \geq \int_{\mathbb{R}} g^* \, \mathrm{d}\mathcal{L}(x) = +\infty$. Hence no integrable dominating profile exists. Here, convergence fails because the sequence concentrates its entire integral into sets of arbitrarily small measure near the origin with unbounded height.

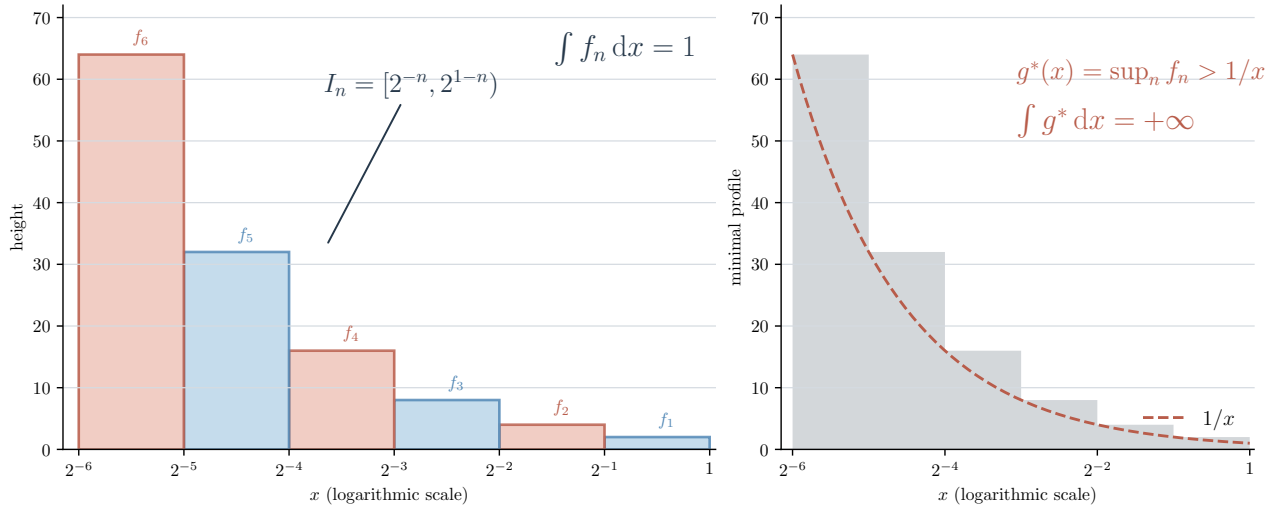


Figure 9.3: The unit mass is packed into intervals shrinking toward zero. The minimal dominating profile has the same non-integrable singularity as $1/x$.

Figure description: Two side-by-side plots show unit-mass rectangular pulses on intervals shrinking toward zero and the resulting stepped minimal profile, which lies above a dashed curve proportional to one over x and becomes unbounded near zero.

These two examples isolate the two mechanisms that disrupt the convergence of integrals: mass spreading over sets of infinite measure (escape to infinity) and mass concentrating into sets of vanishing measure with unbounded amplitude (concentration). An integrable dominator g prevents both behaviors simultaneously: by Markov's inequality, g has finite-measure superlevel sets (ruling out escape to infinity), and by the uniform integrability of integrable functions, g is uniformly integrable, meaning its integral on sets of small measure is uniformly small (ruling out concentration).

9.2 A proof using Fatou's lemma

The monotone convergence theorem requires the sequence (f_n) to be non-decreasing. In practice, however, sequences of functions often oscillate and may fail to be monotone, or even fail to converge pointwise. Even when a pointwise limit does not exist, the limit inferior always exists in $[0, +\infty]$ for every sequence of non-negative measurable functions. Fatou's lemma provides a fundamental one-sided inequality relating the integral of the limit inferior to the limit inferior of the integrals, with no monotonicity assumptions.

For a sequence of extended real numbers $(a_n)_{n\in\mathbb{N}}$ in $[-\infty, +\infty]$, the **limit inferior** is defined by

$$\liminf_{n\to\infty} a_n \coloneqq \lim_{n\to\infty} \left(\inf_{k\geq n} a_k \right).$$

Because the tail infima $t_n \coloneqq \inf_{k\geq n} a_k$ form an increasing sequence $(t_n \leq t_{n+1})$, their limit always exists in $[-\infty, +\infty]$.

Theorem 9.5 (Fatou's lemma). Let $(h_n)_{n\in\mathbb{N}}$ be a sequence of measurable functions $h_n \colon X \to [0, +\infty]$.

Then

$$\int_X \liminf_{n\to\infty} h_n \, \mathrm{d}\mu \le \liminf_{n\to\infty} \int_X h_n \, \mathrm{d}\mu. \tag{9.3}$$

Fatou's lemma is proved as follows. For each $n \in \mathbb{N}$, define $q_n(x) \coloneqq \inf_{k\ge n} h_k(x)$. Each q_n is measurable as a countable infimum of measurable functions, and $0 \le q_n \le q_{n+1}$. By definition of the limit inferior, $q_n \uparrow \liminf_{n\to\infty} h_n$ pointwise on X. Applying the monotone convergence theorem to (q_n) yields

$$\int_X \liminf_{n\to\infty} h_n \, \mathrm{d}\mu = \lim_{n\to\infty} \int_X q_n \, \mathrm{d}\mu.$$

On the other hand, for every $j \ge n$, $q_n(x) \le h_j(x)$ everywhere on X, so monotonicity gives

$$\int_X q_n \, \mathrm{d}\mu \le \int_X h_j \, \mathrm{d}\mu \qquad \text{for all } j \ge n.$$

Taking the infimum over $j \ge n$ on the right-hand side yields $\int_X q_n \, \mathrm{d}\mu \le \inf_{j\ge n} \int_X h_j \, \mathrm{d}\mu$. Taking $n \to \infty$ on both sides establishes (9.3).

First proof of Theorem 9.1. Let N_0 be the null set on which the sequence $(f_n(x))$ fails to converge. Define the pointwise limit $f(x) \coloneqq \lim_{n\to\infty} f_n(x)$ for $x \in X \setminus N_0$, and set $f(x) \coloneqq 0$ for $x \in N_0$. As a pointwise limit of measurable functions outside a null set, f is measurable. For each $n \in \mathbb{N}$, let N_n be the null set where $|f_n(x)| > g(x)$. Setting $N \coloneqq \bigcup_{n=0}^{\infty} N_n$, we have $\mu(N) = 0$ by countable subadditivity. Replacing each function with 0 on N preserves measurability and leaves all integrals unchanged. We may therefore assume without loss of generality that $f_n(x) \to f(x)$ and $|f_n(x)| \le g(x)$ for every $x \in X$ and all $n \in \mathbb{N}$. In particular, taking the limit as $n \to \infty$ shows that $|f(x)| \le g(x)$ everywhere on X, so f and every f_n are integrable since their absolute values have integrals bounded by $\int_X g \, \mathrm{d}\mu < +\infty$.

Step 1: lower bound via Fatou's lemma.
Because $|f_n| \le g$, the functions $g + f_n$ are non-negative and measurable. Pointwise, $\liminf_{n\to\infty}(g + f_n) = g + f$. Applying Fatou's lemma to $(g + f_n)$ yields

$$\int_X (g + f) \, \mathrm{d}\mu \le \liminf_{n\to\infty} \int_X (g + f_n) \, \mathrm{d}\mu.$$

By linearity of the integral for integrable functions, $\int_X (g + f_n) \, \mathrm{d}\mu = \int_X g \, \mathrm{d}\mu + \int_X f_n \, \mathrm{d}\mu$. Because $\int_X g \, \mathrm{d}\mu < +\infty$, we subtract it from both sides to obtain

$$\int_X f \, \mathrm{d}\mu \le \liminf_{n\to\infty} \int_X f_n \, \mathrm{d}\mu. \tag{9.4}$$

Step 2: upper bound via Fatou's lemma.
Similarly, the functions $g - f_n$ are non-negative and measurable, with $\liminf_{n\to\infty}(g - f_n) = g - f$. Fatou's lemma gives

$$\int_X (g - f) \, \mathrm{d}\mu \le \liminf_{n\to\infty} \int_X (g - f_n) \, \mathrm{d}\mu = \int_X g \, \mathrm{d}\mu - \limsup_{n\to\infty} \int_X f_n \, \mathrm{d}\mu.$$

Subtracting the finite quantity $\int_X g \, \mathrm{d}\mu$ and multiplying by -1 reverses the inequality:

$$\limsup_{n\to\infty} \int_X f_n \, \mathrm{d}\mu \le \int_X f \, \mathrm{d}\mu. \tag{9.5}$$

Combining (9.4) and (9.5) gives

$$\lim_{n\to\infty} \int_X f_n \, \mathrm{d}\mu = \int_X f \, \mathrm{d}\mu = \int_X \lim_{n\to\infty} f_n \, \mathrm{d}\mu,$$

which proves (9.1).

Step 3: convergence in L^1.
To establish the stronger conclusion (9.2), consider the functions $u_n := |f_n - f|$. The sequence (u_n) converges pointwise to 0 on X and satisfies the domination bound

$$|u_n(x)| \le |f_n(x)| + |f(x)| \le 2g(x) \qquad \text{for all } x \in X.$$

Because $2g$ is integrable, the integral convergence established in Steps 1 and 2 applies directly to (u_n), giving

$$\lim_{n\to\infty} \int_X |f_n - f| \, \mathrm{d}\mu = \int_X 0 \, \mathrm{d}\mu = 0.$$

This proves (9.2). □

9.3 Vitali convergence via almost uniform convergence

In Proposition 9.2, the proof of dominated convergence for series rested on two observations:

1. the sum outside a finite set $\{1, \dots, K\}$ is small because $\sum_{k=K+1}^{\infty} b(k)$ is a convergent tail;
2. on the finite set $\{1, \dots, K\}$, pointwise convergence $a_n(k) \to a(k)$ is automatically uniform.

On a general measure space $(X, \mathcal{A}, \mu)$, the first mechanism translates naturally: by Markov's inequality, an integrable dominator g is supported "mostly" on a set of finite measure $K = \{g > 2^{-m}\}$, with the tail integral $\int_{X\setminus K} g \, \mathrm{d}\mu$ arbitrarily small.

However, the second mechanism encounters an obstacle: on a general measure space, pointwise convergence almost everywhere does *not* imply uniform convergence, even when the ambient space has finite measure! To address this issue we prove Egorov's theorem, which shows that on finite measure spaces pointwise convergence implies uniform convergence up to sets of arbitrarily small measue.

9.3.1 Egorov's theorem

Theorem 9.6 (Egorov's theorem). Let $(X, \mathcal{A}, \mu)$ be a measure space with $\mu(X) < +\infty$. Suppose measurable functions $f_n \colon X \to \mathbb{R}$ converge μ-almost everywhere to a measurable function $f \colon X \to \mathbb{R}$. For every $\eta > 0$, there exists a measurable set $E \subseteq X$ such that $\mu(E) < \eta$ and $f_n \to f$ uniformly on $X \setminus E$.

Proof of Theorem 9.6. Let N be the null set on which $f_n(x) \not\to f(x)$. For integers $j \ge 1$ and $n \ge 1$, define the set of points where some term beyond index n differs from the limit by at least 2^{-j}:

$$E_{j,n} := \bigcup_{k \ge n} \left\{ x \in X \setminus N \colon |f_k(x) - f(x)| \ge 2^{-j} \right\}.$$

For each fixed scale j, the sets $E_{j,n}$ are nested: $E_{j,n+1} \subseteq E_{j,n}$. Because $f_k(x) \to f(x)$ for every $x \in X \setminus N$, every such x eventually satisfies $|f_k(x) - f(x)| < 2^{-j}$ for all sufficiently large k. Therefore

$$\bigcap_{n=1}^{+\infty} E_{j,n} = \emptyset.$$

Because $\mu(X) < +\infty$, the measure of the initial set is finite: $\mu(E_{j,1}) \le \mu(X) < +\infty$. Continuity of

measure from above implies

$$\lim_{n\to\infty} \mu(E_{j,n}) = 0.$$

Let $\eta > 0$. For each integer $j \geq 1$, choose an index $n_j \in \mathbb{N}$ such that

$$\mu(E_{j,n_j}) < \frac{\eta}{2^{j+1}}.$$

Define the exceptional set

$$E \coloneqq N \cup \bigcup_{j=1}^{+\infty} E_{j,n_j}.$$

By countable subadditivity of the measure,

$$\mu(E) \leq \mu(N) + \sum_{j=1}^{+\infty} \mu(E_{j,n_j}) < 0 + \sum_{j=1}^{+\infty} \frac{\eta}{2^{j+1}} = \frac{\eta}{2} < \eta.$$

It remains to verify uniform convergence on $X \setminus E$. If $x \in X \setminus E$, then for every $j \geq 1$, $x \notin E_{j,n_j}$. By definition of E_{j,n_j}, this means

$$|f_k(x) - f(x)| < 2^{-j} \qquad \text{for all } k \geq n_j.$$

Let $\varepsilon > 0$. Choose an integer $j \geq 1$ such that $2^{-j} < \varepsilon$. Then for all $k \geq n_j$,

$$\sup_{x\in X\setminus E} |f_k(x) - f(x)| \leq 2^{-j} < \varepsilon.$$

This proves that $f_k \to f$ uniformly on $X \setminus E$. □

Example 9.7 (Failure of Egorov's theorem on an infinite-measure space). Let $X = \mathbb{N}$ equipped with counting measure, so $\mu(\mathbb{N}) = +\infty$, and define

$$f_n(k) \coloneqq \mathbb{1}_{\{k\geq n\}}.$$

For each fixed $k \in \mathbb{N}$, $f_n(k) = 0$ for all $n > k$, so $f_n(k) \to 0$ pointwise everywhere. However, for every n, $\sup_{k\in\mathbb{N}} f_n(k) = 1$, so the convergence is not uniform on $\mathbb{N}$. If $\eta = 1/2$, any set $E \subseteq \mathbb{N}$ with $\mu(E) < \eta$ must have $\mu(E) = 0$, which implies $E = \emptyset$. Thus $X \setminus E = \mathbb{N}$, where convergence fails to be uniform. This demonstrates that the finite-measure hypothesis $\mu(X) < +\infty$ in Egorov's theorem cannot be omitted.

9.3.2 Tightness and equi-uniform integrability

Instead of proving Theorem 9.1, we prove a more general theorem (Vitali). To formulate the theorem we introduce two properties of familities of functions. These properties are automatically satisfied for any fixed integrable function.

Definition 9.8 (Tightness). A family of integrable functions $\mathcal{F} \coloneqq \{f\colon X \to \mathbb{R}\}$ is **tight** if for every $\varepsilon > 0$ there exists a measurable set $K \subseteq X$ such that

$$\mu(K) < +\infty \quad \text{and} \quad \int_{X\setminus K} |f| \, \mathrm{d}\mu < \varepsilon \quad \forall f \in \mathcal{F}.$$

The same set K must work for every function $f \in \mathcal{F}$. If $\mu(X) < +\infty$, every sequence of integrable functions is tight, by taking $K = X$.

Definition 9.9 (Equi-uniform integrability). A family $\mathcal{F}$ of integrable functions $f\colon X \to \mathbb{R}$ is **equi-uniformly integrable** if for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$A \in \mathcal{A} \quad \text{and} \quad \mu(A) < \delta \implies \int_A |f| \, \mathrm{d}\mu < \varepsilon \quad \forall f \in \mathcal{F}.$$

Unlike uniform integrability of an individual function, this condition requires one choice of δ to work for the entire family of function. Tightness rules out escape of mass, while equi-uniform integrability rules out concentration. For instance, the translates in Example 9.3 are equi-uniformly integrable because $|f_n| \le 1$, but are not tight: for any finite-measure set K, the disjointness of $[n, n+1)$ implies $\mathcal{L}(K \cap [n, n+1)) \to 0$, so $\int_{\mathbb{R}\setminus K} |f_n| \, \mathrm{d}\mathcal{L}(x) \to 1$. The concentrating sequence in Example 9.4 is tight, since every term is supported in $[0, 1]$, but is not equi-uniformly integrable: $\mathcal{L}(I_n) = 2^{-n} \to 0$ while $\int_{I_n} |f_n| \, \mathrm{d}\mathcal{L}(x) = 1$.

9.3.3 Vitali's convergence theorem

Theorem 9.10 (Vitali's convergence theorem). Let $f_n\colon X \to \mathbb{R}$ be integrable functions that converge μ-almost everywhere to a measurable function $f\colon X \to \mathbb{R}$. If (f_n) is tight and equi-uniformly integrable, then f is integrable and

$$\lim_{n\to\infty} \int_X |f_n - f| \, \mathrm{d}\mu = 0.$$

In particular,

$$\lim_{n\to\infty} \int_X f_n \, \mathrm{d}\mu = \int_X f \, \mathrm{d}\mu.$$

Proof of Theorem 9.10. After setting the functions equal to zero on a measurable null set, we may assume $f_n \to f$ everywhere. Let $\varepsilon > 0$.

Step 1: control the tails and small sets, including for the limit.
By tightness, choose a measurable set K of finite measure such that

$$\sup_{n\in\mathbb{N}} \int_{X\setminus K} |f_n| \, \mathrm{d}\mu < \varepsilon. \tag{9.6}$$

By equi-uniform integrability, choose $\delta > 0$ such that

$$A \in \mathcal{A} \quad \text{and} \quad \mu(A) < \delta \implies \sup_{n\in\mathbb{N}} \int_A |f_n| \, \mathrm{d}\mu < \varepsilon. \tag{9.7}$$

For every measurable set A, Fatou's lemma gives

$$\int_A |f| \, \mathrm{d}\mu \le \liminf_{n\to\infty} \int_A |f_n| \, \mathrm{d}\mu \le \sup_{n\in\mathbb{N}} \int_A |f_n| \, \mathrm{d}\mu.$$

Thus the same tail and small-set bounds hold for f, even though we have not yet established its integrability on all of X.

Step 2: apply Egorov's theorem on the finite-measure set.
By Theorem 9.6, there exists a measurable set $E \subseteq K$ with $\mu(E) < \delta$ such that $f_n \to f$ uniformly on $K \setminus E$. In particular, for some index n_0 we have $|f - f_{n_0}| \le 1$ on $K \setminus E$. Consequently,

$$\int_{K\setminus E} |f| \, \mathrm{d}\mu \le \int_X |f_{n_0}| \, \mathrm{d}\mu + \mu(K) < +\infty.$$

The bounds from Step 1 give $\int_E |f|\,\mathrm{d}\mu < \varepsilon$ and $\int_{X\setminus K} |f|\,\mathrm{d}\mu < \varepsilon$. Together these show that f is integrable on X.

Step 3: estimate the three regions.
Uniform convergence provides an index N such that for every $n \geq N$ and every $x \in K \setminus E$,

$$|f_n(x) - f(x)| < \frac{\varepsilon}{\mu(K)+1}.$$

Decomposing $X = (K \setminus E) \cup E \cup (X \setminus K)$, we obtain, for every $n \geq N$,

$$\begin{aligned}\int_X |f_n - f|\,\mathrm{d}\mu &\leq \frac{\varepsilon\mu(K)}{\mu(K)+1} + \int_E |f_n|\,\mathrm{d}\mu + \int_E |f|\,\mathrm{d}\mu \\ &\quad + \int_{X\setminus K} |f_n|\,\mathrm{d}\mu + \int_{X\setminus K} |f|\,\mathrm{d}\mu \\ &< \varepsilon + 2\varepsilon + 2\varepsilon = 5\varepsilon.\end{aligned}$$

Since $\varepsilon > 0$ was arbitrary, this proves convergence in L^1. Finally,

$$\left|\int_X f_n\,\mathrm{d}\mu - \int_X f\,\mathrm{d}\mu\right| \leq \int_X |f_n - f|\,\mathrm{d}\mu \longrightarrow 0.$$

□

9.3.4 Dominated convergence as a consequence

Second proof of Theorem 9.1. Suppose $f_n \to f$ almost everywhere and $|f_n| \leq g$ for an integrable function $g \geq 0$. Every f_n is integrable. We verify the two hypotheses of Theorem 9.10.

Tightness.
As $m \to \infty$, the functions $g\mathbb{1}_{\{g>2^{-m}\}}$ increase pointwise to g. By monotone convergence and the finiteness of $\int_X g\,\mathrm{d}\mu$,

$$\int_{\{g\leq 2^{-m}\}} g\,\mathrm{d}\mu \longrightarrow 0.$$

Given $\varepsilon > 0$, choose m so this integral is less than ε, and set $K = \{g > 2^{-m}\}$. Markov's inequality gives

$$\mu(K) \leq 2^m \int_X g\,\mathrm{d}\mu < +\infty, \qquad \sup_{n\in\mathbb{N}} \int_{X\setminus K} |f_n|\,\mathrm{d}\mu \leq \int_{X\setminus K} g\,\mathrm{d}\mu < \varepsilon.$$

Thus (f_n) is tight.

Equi-uniform integrability.
Because g is integrable, for every $\varepsilon > 0$ there exists $\delta > 0$ such that $\mu(A) < \delta$ implies $\int_A g\,\mathrm{d}\mu < \varepsilon$. For every such measurable set A,

$$\sup_{n\in\mathbb{N}} \int_A |f_n|\,\mathrm{d}\mu \leq \int_A g\,\mathrm{d}\mu < \varepsilon.$$

Thus (f_n) is equi-uniformly integrable. Vitali's theorem now yields the integrability of f, convergence in L^1, and convergence of the integrals, proving both conclusions of dominated convergence. □